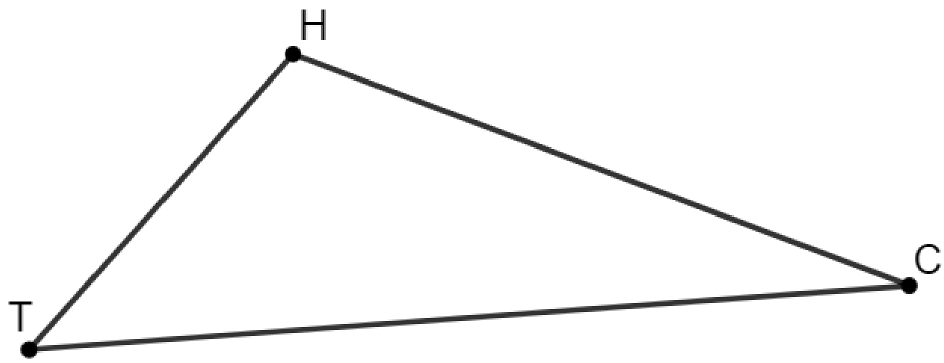
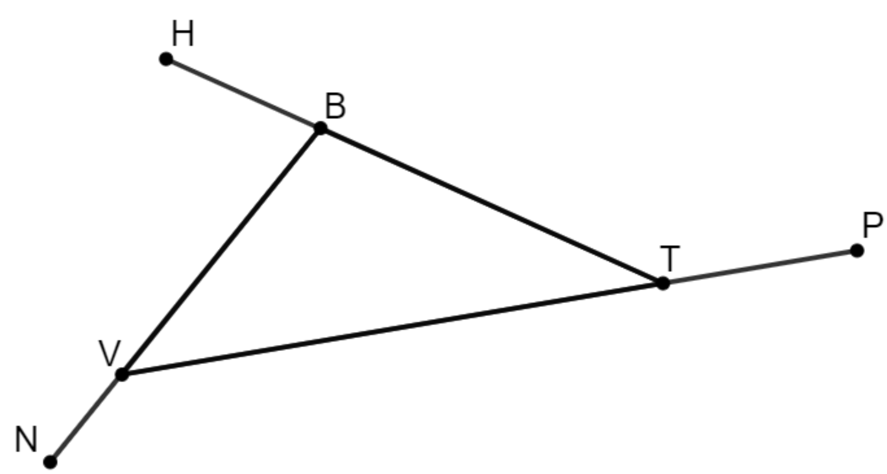
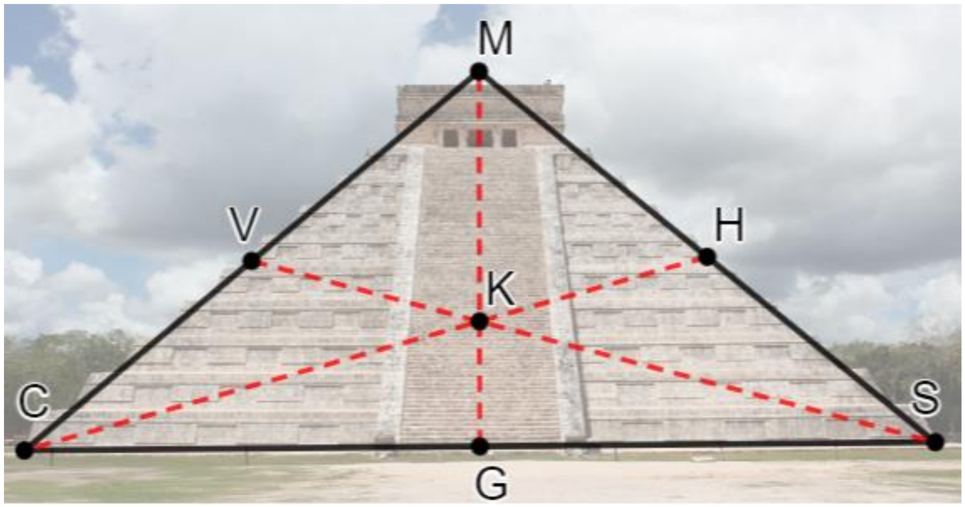
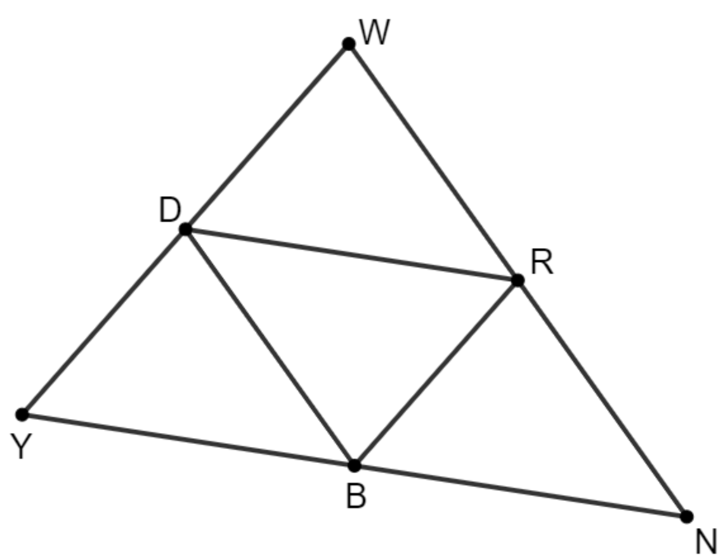
11.9.2 Activity

Problem A
ΔTHC is shown, where $m\angle THC = (9x - 15)°$, $m\angle HCT = (x + 11)°$, and $m\angle CTH = (3x + 2)°$.  What is the measure of $\angle THC$?

Problem B
Jonah is creating a triangular garden. He has three pieces of fencing that measure 5 feet (ft), 5 ft, and 11 ft. Does Jonah have enough fencing to make a triangular garden? Justify your answer using math.

Problem C
ΔVBT is shown, where $\angle VBH$, $\angle NVT$, and $\angle PTB$ are exterior angles. $m\angle VTB = 34°$, $m\angle NVT = (18x - 14)°$, and $m\angle VBH = (6x + 24)°$.  What is the measure of $\angle NVT$?

Problem D
A picture of El Castillo Pyramid is shown with ΔCMS drawn on the picture. Points V, H, and G are midpoints of each side of the triangle. The medians $\overline{GM}$, $\overline{CH}$, and $\overline{SV}$ are shown. $MK = (8x + 2)$ meters (m) and $KG = (\frac{4}{3}x + 7)$ m.  If Kyra is standing at point K, how far did she walk up the stairs from point G?

Problem E
ΔWNY is shown, where points D, R, and B are midpoints of each side of the triangle. $RB = 11$, $YN = (5x)$, $DR = (\frac{1}{2}x + 12)$, $NW = (5z + 1)$, and $DB = (3z - 2)$.  What is the perimeter of ΔWNY?

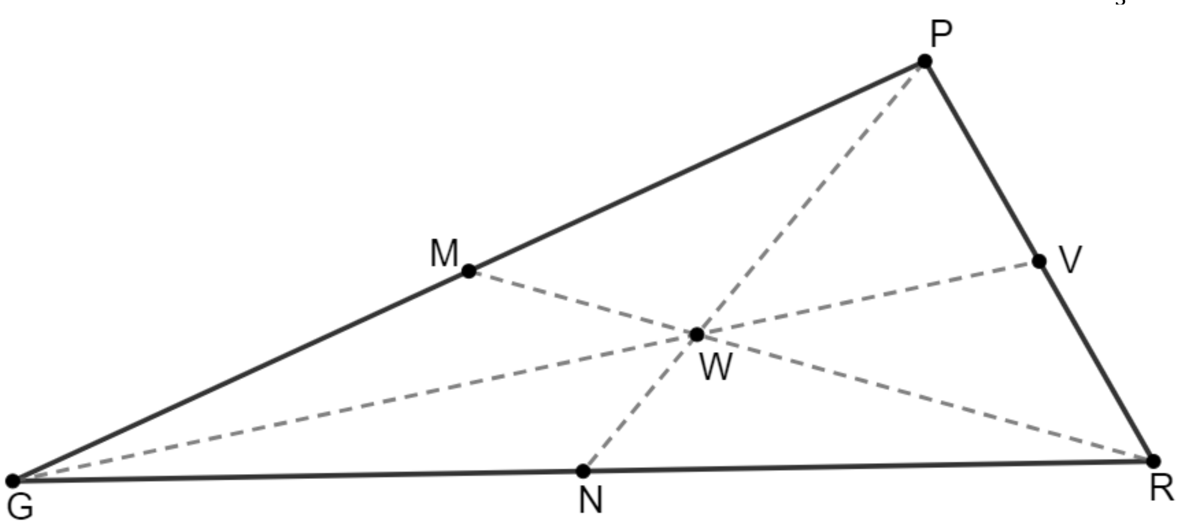
Problem F

The roof of the building creates ΔSWD, where $m\angle WSD = (3x + 7)°$, $m\angle SWD = (9x + 1)°$, and $m\angle WDS = (5x - 15)°$.

What is the measure of angle $m\angle SWD$?

Problem G

ΔPRG is shown, where points M, V, and N are the midpoints of each side of the triangle and the perimeter of ΔPRG is 89. $GM = (4x - 1)$, $GR = (8x + 3)$, and $VR = (\frac{4}{3}x + 2)$.



What is the value of x?